

Toran Bahadur Niroula | Aerospace Engineering Student

Pulchowk, Lalitpur, Nepal

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🔗 Toran999 • 📅 Expected Graduation: June 2026

🎓 Education

Institute of Engineering, Pulchowk Campus

Lalitpur, Nepal

Bachelor of Engineering in Aerospace Engineering

2022–2026

- **Cumulative Percentage:** 82%
- **Academic Standing:** Top 1% of class overall
- **Relevant Coursework:** Fluid Dynamics, Control Systems, Numerical Methods, CFD, Aircraft Structures, Flight Dynamics, UAV Design, Aircraft Preliminary Design, Compressible Aerodynamics

🛠 Experience

IIT Kanpur

Kanpur, India

Intern, Experimental Aerodynamics Group

Oct–Nov 2025

- Processed and analyzed PIV data using **PIVlab**; extracted velocity, vorticity, and Reynolds stress fields
- Performed **POD** and modal analysis in MATLAB/Python to study coherent structures
- Gained exposure to tufts visualization and aerodynamic force/torque measurement techniques

FOSSEE, IIT Bombay

Remote

CFD Research Intern

Feb–Jun 2025

- Conducted aerodynamic simulations of fixed-wing UAV configurations using OpenFOAM
- Analyzed stall characteristics and performance benefits of blended winglet designs
- Performed grid independence studies and turbulence model validation
- Prepared comprehensive technical reports documenting methodology and findings

DBF Pulchowk (AIAA Design/Build/Fly)

IOE Pulchowk

Design Team Member

2023–2024

- Contributed to development of competition aircraft *Junkiri*
- Collaborated on aerodynamic design, structural optimization, and flight testing
- Contributed to the preparation of technical documentation for competition submission

🔧 Projects

Study of Surface Heating Effects on Compressible Wall jets using Rainbow Schlieren Deflectometry:

Final Year Project

2026

- Investigating thermal boundary layer interactions at high subsonic speeds
- Implementing schlieren visualization techniques for flow analysis
- Development of Quantitative Schlieren Facility and post processing packages for Rainbow Schlieren Deflectometry.

Electric Agricultural UAV Conceptual Design: *Design Study*

2025

- Conceptualized an electric crop-dusting UAV of 200 kg payload capacity and 45-minute endurance
- Conducted aerodynamic analysis using XFLR5 and performance simulations in X-Plane 12
- Presented final design through a technical poster showcasing performance specifications

Twin-Boom UAV Design and Fabrication: *Team Project*

2025

- Designed and fabricated a twin-boom fixed-wing UAV with a 3-member team
- Used **XFLR5** for aerodynamic analysis and **hot-wire foam cutting** for surface fabrication
- Achieved stable and successful flight tests validating design performance

Technical Skills

CFD & Simulation: OpenFOAM, ANSYS Fluent, XFLR5, **ParaView**, PIVlab

Experimental Analysis: Schlieren, Rainbow Schlieren Deflectometry, PIV

Data Analysis: Python (NumPy, Matplotlib), MATLAB, POD/Modal Analysis

CAD & Design: CATIA, SolidWorks, X-Plane

Documentation: LaTeX

Trainings & Certifications

2025: Samsung Innovation Campus: Coding and Programming Certification

2023: edX: Python Basics for Data Science

2023: edX: Using Python for Research

Interests

Computational Fluid Dynamics, Experimental Aerodynamics, Optical Flow Diagnostics, Schlieren, PIV, Data-Driven Flow Analysis, UAV Design